When two compound propositions always have the same truth values, regardless of the truth values of its propositional variables, we call them $\boxed{\text{equivalent}}$.

- Show that a conditional statement and its contrapositive are logically equivalent.

p	q	$p \to q$	$\neg p$	$\neg q$	$\neg q \to \neg p$
T	T	T	F	F	T
T	F	F	F	T	F
F	T	T	T	F	T
F	F	T	T	T	T

- Show that the converse and inverse of a conditional statement are logically equivalent.

p	q	$q \to p$	$\neg p$	$\neg q$	$\neg p \to \neg q$
T	T	T	F	F	T
T	F	T	F	T	T
F	T	F	T	F	F
F	F	T	T	T	T

$\boxed{\text{Example 12:}}$ Find the contrapositive, the converse and the inverse of the conditional statement "The home team wins whenever it is raining" ✓

$\boxed{\text{Soln:}}$ Conditional ⟶ "If it is raining, then the home team wins" ✓

Converse ⟶ "If the home team wins, then it is raining."

Inverse ⟶ "If it is not raining, then the home team will not win." ✓

Contrapositive ⟶ "If the home team does not win, then it is not raining."

$\boxed{\text{Definition 6:}}$ Let p and q be propositions. The $\boxed{\text{biconditional statement}}$ $p \leftrightarrow q$ is the proposition "p if and only if q". The biconditional statement $p \leftrightarrow q$ is true when p and q have the same truth values, and is false otherwise. Biconditional statements